BSc (HONS.) IN COMPUTER SCIENCE AND ENGINEERING

FIRST YEAR, SECOND SEMESTER EXAMINATION, 2020

[According to the New Syllabus]

LINEAR ALGEBRA

Subject Code: CSE-510225

Examination Code : 5612

Time-3 hours

Full marks--80

(N.B. The figures in the right margin indicate full marks. Answer any four questions.)

1. (a) Define homogeneous and non-homogeneous linear equation. 3+5=8
Reduce the following system of linear equations into echelon form and solve it.

$$\begin{aligned} x_1 - x_2 + 2x_3 &= 5 \\ 2x_1 + x_2 - x_3 &= 2 \\ 2x_1 - x_2 - x_3 &= 4 \\ x_1 + 3x_2 + 2x_3 &= 1 \end{aligned}$$

(b) Determine the values of λ such that the following system of linear equations in unknowns x, y and z has (i) a unique solution (ii) no solution (iii) more than one solution :

$$\begin{aligned} x + y + \lambda z &= 1 \\ x + \lambda y + z &= \lambda \\ \lambda x + y + z &= \lambda^2 \end{aligned}$$

(c) If $2s = a + b + c$ then prove that,

$$\begin{vmatrix} a^2 & (s-a)^2 & (s-a)^2 \\ (s-b)^2 & b^2 & (s-b)^2 \\ (s-c)^2 & (s-c)^2 & c^2 \end{vmatrix}$$

2. (a) Define : Singular Matrix, Nilpotent Matrix and Hermitical Matrix 2x3=6.
(b) If A and B are non-singular matrices, then prove that, 7

$$(AB)^{-1} = B^2A^2$$

(c) Define the rank of a matrix. Find the rank of the following matrix : 2+5=7

$$\begin{bmatrix} 1 & 2 & 0 & -1 \\ 2 & 6 & -3 & -3 \\ 3 & 10 & -6 & -5 \end{bmatrix}$$

3. (a) If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ prove that, $A^2 = A^{-1}$

(b) Show that the matrix $A = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$

is invertible for all values θ and find A^{-1}.

4. (a) Define linear transformation let T: 13 ~>R3 be given by $T(a, b, c)$ _(a, b, o) $\forall a, b, c \in R$. . Show that T is a linear transformation. 2+5=7

(b) Define rank and nullity of a linear • transformation. If $T: U \rightarrow V$ is a linear transformation then show that rank of $T + nullity\ of\ T = dim\ U$ 2+6=8

(c) $T: R^3 \rightarrow R^3$ be the linear operator defined by $T(x, y, z) = (x + 2y - z, 2x - y + 4z, 4x + 3y - 2z)$. Find a basis and the dimension of the (i) Im T of T (ii) K.er T of T.

5. (a) Define matrix representation of a linear operator. Let $\{u_1,, u_2, ... U_m\}$ be a basis of Uand T be any linear operator of U, then for any vector $W \in U$, Show that $[T]_u\ [W]_u = [T\ (W)]_u$. 10

(b) The mapping $T: R^3 \rightarrow R^3$ is defined by $T\ (x, y, z)$ _$(2y + z, x - 4)), 3x)$. Show that, T is a linear transformation and find the matrix T in the basis $\{u_1 =$ _$(1, 1, 1), u_2 = (1, 1, 0), u_3 = (1, 0, 0)\}$. 10

6. (a) Define Eigen values and Eigen vectors. State and prove the Clayey -Hamilton theorem. 2+6=8

(b) Find the Eigen values and the corresponding Eigen vectors of the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix}$ 12

Also verify Cayley-Hamilton theorem for the matrix A.

B.Sc (HONS.) IN COMPUTER SCIENCE AND ENGINEERING FIRST YEAR SECOND SEMESTER EXAM -ATION, 2019

[According to the New Syllabus

CSE-510225

(Linear Algebra)

Time-3 hours

Full marks-80

[N.B. The, figures fn the right margin indicate full marks. Answer any four questions.]

Marks

1. (a) Define linear equation with examples. 2

(b) Determine the value of k such that the following system of linear equations in unknowns x, y, z has : 6

(i) a unique solution (ii) no solution (iii) more than one solutions.

(c) Define norm of a vector. Consider the points, P (3, k, -2) and Q(5, 3, 4) in R^3 . Find the value of λ so that $\overrightarrow{PQ}$ is orthogonal to the vector (4, -3, 2). 6

(d) Prove that, $\begin{bmatrix} -a & -b & c & d \\ b & -a & -d & c \\ c & -d & a & -b \\ d & c & b & a \end{bmatrix} = (a^2 + b^2 + c^2 + a^2)^2$. 6

2. (a) Define (i) symmetric matrix (ii) square matrix (iii) idempotent matrix (iv) singular matrix. 4

(b) If A and B are comparable matrices and AT and BT are the transpose matrices of A and B respectively, then prove that 1+1+4 =6

(i) (AT)T =A, (ii) (A + B)T =AT ± BT, (iii) (AB)T = BTAT.

(c) Solve the following system of linear equations with the help of matrix : 5

x+2y+3z+4=0

2x+4y+5z+7=0

3x+5y+6z+ 10=0

(d) Show that the matrix $A=\begin{bmatrix} 2 & -1 & 1 \\ -2 & 3 & -2 \\ -4 & 4 & -3 \end{bmatrix}$ is idempoterrt. 5

3. (a) Define linear transformation. Show that the product of two linear-transformation is a linear transformation. 2+5=7

(b) Let S and T be the linear operators of RZ into RZ defined by S(x,Y)=3x+2y,-6x+y) T(x,Y)=(2x+y,x-y). Find formulae defining properties S+ 7; ST, TS and S2. 7

(c) Define (i) System of linear equation 6

(ii) Consistent

(iii) Homogeneous

(iv) Non-homogeneous

4. (a) Define minors and cc-factors with example. 3

(b) If $\Delta = \begin{vmatrix} x & x^2 & x^3+1 \\ y & y^2 & y^3+1 \\ z & z^2 & z^3+1 \end{vmatrix}$ I then prove that, 6

A=(x -y) (y -z) (z -x) (xyz - 1). Also show that if x, y and z are not equal and A = 0, then xyz + 1- 0.

(c) Prove that,

$$\begin{bmatrix} 1 & a & a^2 & a^3 \\ 1 & b & b^2 & b^3 \\ 1 & c & c^2 & c^3 \\ 1 & d & d^2 & d^3 \end{bmatrix}$$

5. (a) Define image and kernel of a linear transformation. 4

(b) Show that, T: R3 -> R2, where T (x, y, z) _(x + y - z, 2x - y+ 2z) is a linear transformation. Find a basis and dimension for ImT =1 1 and ker T. 1 1

(c) Find a linear transformation T : 1t3 _> 1};4 whose ImT is generated by I(1, 2,0,-4),(2, 0,-], -3)}. 5

6. (a) Define characteristic matrix and characteristic equation. 3

(b) Find the eigen values and eigen vectors of the matrix 8

$A=\begin{bmatrix} 1 & 2 & 2 \\ 1 & 2 & -1 \\ -1 & 1 & 4 \end{bmatrix}$ 4

(c) State Cayley-Hamilton theorem. Using this theorem find the 2+7

inverse of the matrix $A = = \begin{bmatrix} 2 & 0 & -1 \\ 5 & 1 & 0 \\ 0 & 1 & 3 \end{bmatrix}$ =9

B. Sc (HONS.) IN CSE PART-I, SECOND SEMESTER EXAMINAT'ION, 2018

[According to the New 3~yllahusJ

CSE-510211

Examination Code : 5612

(Linear Algebra)

Time-3 hours

Full marks-80

1. (a) Define : Singular 'matrix, Nilpotent matrix and Hermitian matrix. 2x3=6

(b) If A and B are non-singular matrices, then prove that, (AB) $= B^{-1}A^{-1}$ Also prove that, $(A^{-1})^{-1} = A$ and $(A^{-1})^{t} = (A^{t})^{-1}$. 7

(c) Define rank of matrix. Find the rank of the following matrix : 2+5=7

$$A = \begin{matrix} 1 & 2 & 0 & -1 \\ 2 & 6 & -3 & -3 \\ 3 & 10 & -6 & -5 \end{matrix}$$

2. (a) Define, inverse matrix. If $A = \begin{matrix} 1 & -1 & 1 \\ 2 & -1 & 0 \\ 1 & 0 & 0 \end{matrix}$ then find $A^{-1} + 2A^{t}$. $(2 + 8 = 10)$

(b) Show then $\begin{vmatrix} -1 & b & c & d \\ a & -1 & c & d \\ a & b & -1 & d \\ a & b & c & -1 \end{vmatrix}$

$= (a + 1)(b + 1))c + 1)(d + 1)\left[1 - \frac{a}{a+1} - \frac{b}{b+1} - \frac{c}{c+1} - \frac{d}{d+1}\right]$

3. (a) Solve the following system of linear equations with the help of matrix: 8

$x + 2y + z = 2$

$2x - y + 2z = -1$

$3x - 4y - 3z = -16$

(b) (i) Solve the system of linear equations : 6

$2x + y - 2z = 10$

$3x + 2y + 2z = 1$

$5x + 4y + 3z = 4$

(ii) Find the non-trivial solution for the system of linear equations: 6

$x + 2y - 3z = 0$

$2x + 5y = 2z = 0$

$3x - y - 4z = 0$

4. (a) Define degenerate and non-degenerate linear equation. 2
(b) Determine the values of λ such that the following system of linear equations in unknowns x, y and z has (i) a unique solution (ii) no solution (iii) more than one solution: 6

$$x + y + \lambda z = 1$$
$$x + \lambda y + z = \lambda$$
$$\lambda x + y + z = \lambda^2$$

(c) Show that, $\begin{vmatrix} a+b+c & a+b & a & a \\ a+b & a+b+c & a & a \\ a & a & a+b+c & a+b \\ a & a & a+b & a+b+c \end{vmatrix}$ 6

$= c^2(4a + 2b + c)(2b + c)$.

(d) Define norm of a vector. Consider P(3, λ-2) and Q(5, 3, 4) in $\mathbb{R}^3$. Find the value of λ so that, $\overrightarrow{PQ}$ is orthogonal to the vector (4, -3, 2). 6

5. (a) Define linear transformation. Let U and V be two vector spaces over the field F and T_1 and T_2 be linear transformations from U into V then prove that, $T_1 + T_2$ is a linear transformation. 2+5=7

(b) If S and T be the linear transformation S, $T: \mathbb{R} \longrightarrow \mathbb{R}^3$ defined by S(x, y, z) =(y, z, x) and T(x, y, z) =(z + y+ z, 0, 0) then find (i) (TS) (1, 0, 1), (ii) (ST) (1, 0, 1) (iii) (S + T) (1, 0, 1). (iv) (S- T) (1, 0, 1). 8

(c) Show that the mapping $T: \mathbb{R}^2 \longrightarrow \mathbb{R}^3$ defined by $T(x_1, x_2) = (x_1 + x_2, x_1 - x_2, x_1)$ is linear.

6. (a) Define eigen values and eigen vectors. State and prove Cayley-Hamilton theorem. 2+6=8

(b) Find the eigen values and the corresponding eigen vectors of the matrix $A = \begin{bmatrix} 3 & 1 & 1 \\ 2 & 4 & 2 \\ 1 & 1 & 3 \end{bmatrix}$ 12

Also verify the Cayley-Hamilton theorem for the matrix A.